

DATA SCIENCE WITH PYTHON INTERNSHIP
SKILL NEXIS

End-to-End Sales & Profit Analysis Case Study

Project Title : Sales Profit & Analysis
Week 4 – Final Project Report

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Table Of Contents

1. Project Overview	4
2. Data Cleaning & Preparation (Python)	6
3. Analysis Methodology (SQL/Pandas)	7
4. Key Performance Metrics	8
5. Visual Insights - Excel & Power BI Dashboards	9
6. Detailed Findings	11
7. Recommendations	12
8. Deliverables & Conclusion	13

About This Report

This document is the final deliverable of the End-to-End Analysis Case Study. It consolidates work carried out in three stages Python-based data cleaning, SQL/Pandas analysis, and Excel/Power BI visualization into a single, readable report intended for both technical review and business stakeholders. Each section builds on the previous one: Sections 1-3 explain the process and methodology, Section 4 presents the headline numbers, Section 5 shows the supporting charts, and Sections 6-8 translate the data into findings, recommendations and next steps.

List of Charts /Figures

S.NO	Title	Page
1	Sales by Year	8
2	Sales by Country	9
3	Profit by Product	9
4	Sales by Segment	10
5	Profit by Segment	10
6	Profit by Discount Band	10

1 . Project Overview

This project is an End-to-End Analysis Case Study covering the complete data analytics workflow-from raw data cleaning to final business reporting. The goal was to analyze a sales dataset containing 700 transaction records across five countries, five customer segments, and six products, and to derive actionable business insights.

Project Scope

- Clean and explore the dataset using Python (Pandas, NumPy).
- Analyze data using SQL and Pandas to uncover sales and profit patterns.
- Visualize results using Excel charts and an interactive Power BI dashboard.
- Compile a final report including charts, findings, and recommendations.

Deliverables

- Complete Code Jupyter Notebooks (Clean_Explore_data.ipynb, Data_analysis_sql_pandas.ipynb)
- Excel Workbook with dashboard and charts (End_to_End_Excel_Dashboards.xlsx)
- Power BI Dashboard (End_to_End_PowerBI_Dashboard.pbix)
- This Final Report (PDF) - charts, findings and recommendations

Dataset Summary

Field	Details
Total Records	700
Segment	Government, Small Business, Enterprise, Midmarket, Channel Partners
Countries	United States, Canada, France, Germany, Mexico
Product	Paseo, VTT, Velo, Amarilla, Montana, Carretera
Time Period	2013-2014
Key Columns	Units Sold, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit

Business Objective

Beyond building dashboards, the underlying business question this project answers is: where is the company making money, where is it losing money, and what should it do next? Every step of the workflow cleaning, analysis, visualization was designed to answer this question with evidence rather than assumption, and the final sections of this report (6-8) turn that evidence into concrete next steps for management.

Tools & Technologies Used

Stage	Tools	Purpose
Data Cleaning	Python (Pandas, NumPy)	Handle missing values, duplicates, type conversion
Analysis	SQL, Python	Group, aggregate and rank sales/profit by dimension
Visualization	Excel	Static charts, KPI cards, pivot-style summaries
Visualization	Power BI	Interactive dashboard with slicers/filters
Reporting	PDF / word	Final consolidated report for stakeholders

2. Data Cleaning & Preparation (Python)

The raw dataset was loaded and cleaned using Python (Pandas) inside Clean_Explore_data.ipynb. This step ensured the data was consistent, accurate, and ready for analysis.

Steps Performed

- Loaded the raw sales dataset and inspected data types, shape, and column names.
- Checked for missing values - the 'Discount Band' column contained blanks for transactions with no discount applied; these were treated as a 'No Discount' category.
- Checked for and removed duplicate records to avoid double-counting sales.
- Converted the 'Date' column to a proper datetime format and derived Month Number, Month Name, and Year fields for time-based analysis.
- Verified numeric fields (Units Sold, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit) were correctly typed as numbers.
- Standardized text fields (Segment, Country, Product, Discount Band) for consistent spelling and casing.
- Exported the cleaned dataset as cleaned_data.csv for use in the SQL/Pandas analysis and Excel/Power BI dashboards.

Outcome

After cleaning, the dataset contained 700 clean, analysis-ready records with no missing critical values and no duplicates, forming a reliable foundation for the rest of the project.

3. Analysis Methodology (SQL/Pandas)

Using Data_analysis_sql_pandas.ipynb, the cleaned data was analyzed with SQL-style queries and Pandas group-by operations to answer key business questions.

Questions Explored

- Which countries generate the highest sales and profit?
- Which products are the most profitable, and which underperform?
- How do sales trend year-over-year (2013 vs 2014)?
- Which customer segment contributes the most and the least-profit?
- What effect does the discount band (None/Low/Medium/High) have on profit?

Approach

- Aggregated Sales and Profit using GROUP BY-style operations on Country, Product, Segment, Year and Discount Band.
- Calculated Profit Margin % (Profit/Sales) to measure efficiency, not just volume.
- Sorted and ranked each dimension to identify top and bottom performers.
- Cross-checked SQL query results against Pandas group by results for accuracy.
Sample SQL Query

4. Key Performance Metrics

The table below summarizes the headline metrics for the full 2013-2014 dataset, consistent across the Excel workbook and the Power BI dashboard.

Metrics	Values
Total Sales	118,726,350.26
Total Margin	16,893,702.26
Profit Margin	14.2%
Total United Sold	~1.13 Million
Total Records	700

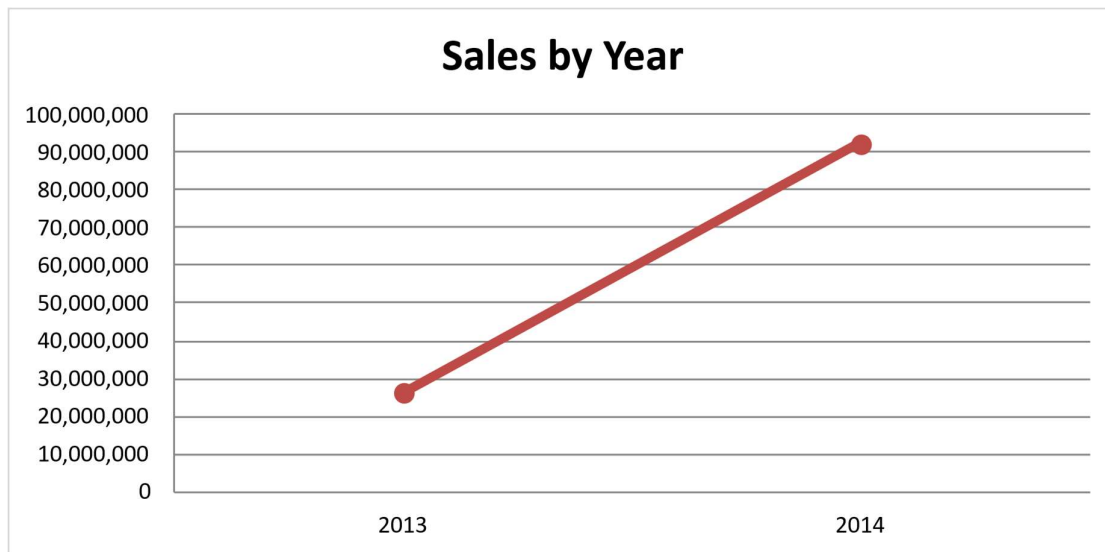


Figure 1

Year-over-Year Breakdown

Year	Sales	Profit	Profit Margin
2013	26,415,255.51	3,878,464.51	14.7%
2014	92,311,094.75	13,015,237.75	14.1%
Growth	+249.5%	+235.6%	-0.6 pts

5. Visual Insights - Excel & Power BI Dashboards

Interactive dashboards were built in both Excel (native charts with slicers) and Power BI (drill-down visuals with filters for Year, Segment, Country and Discount Band). The charts below summarize the same underlying data.

Sales by Country

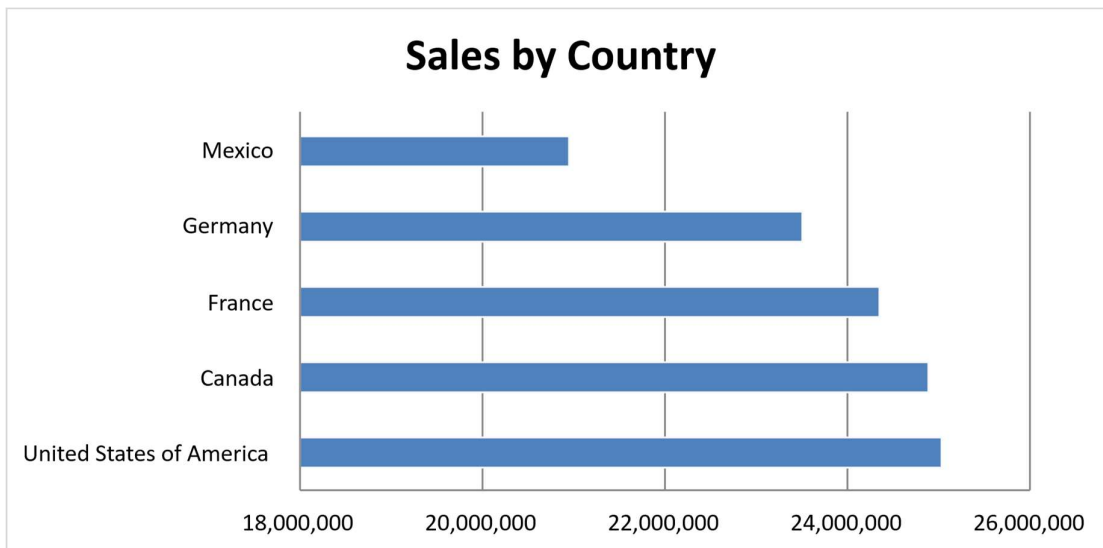


Figure 2

Profit by Product

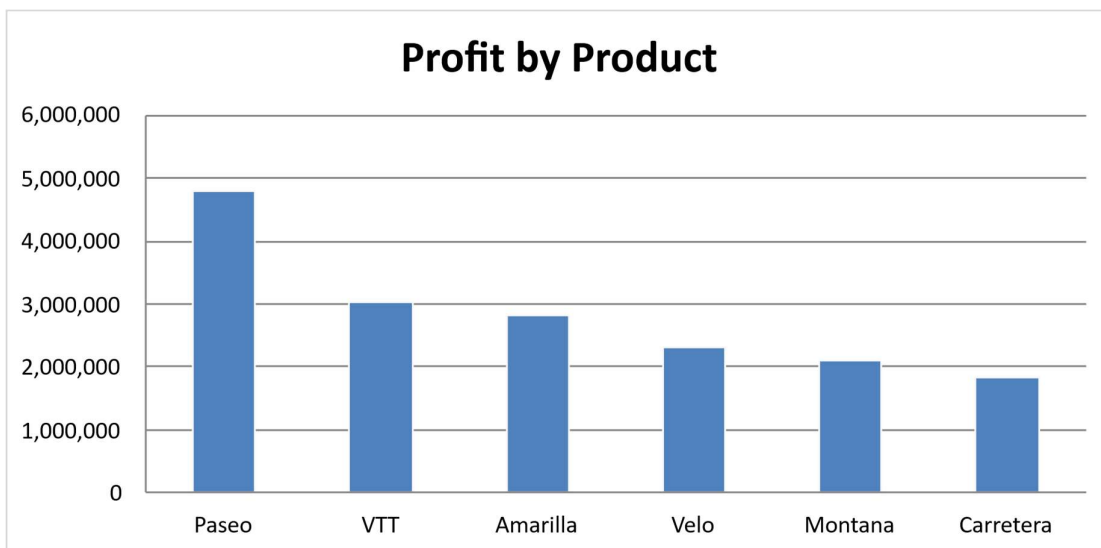


Figure 3

Sales by Segment

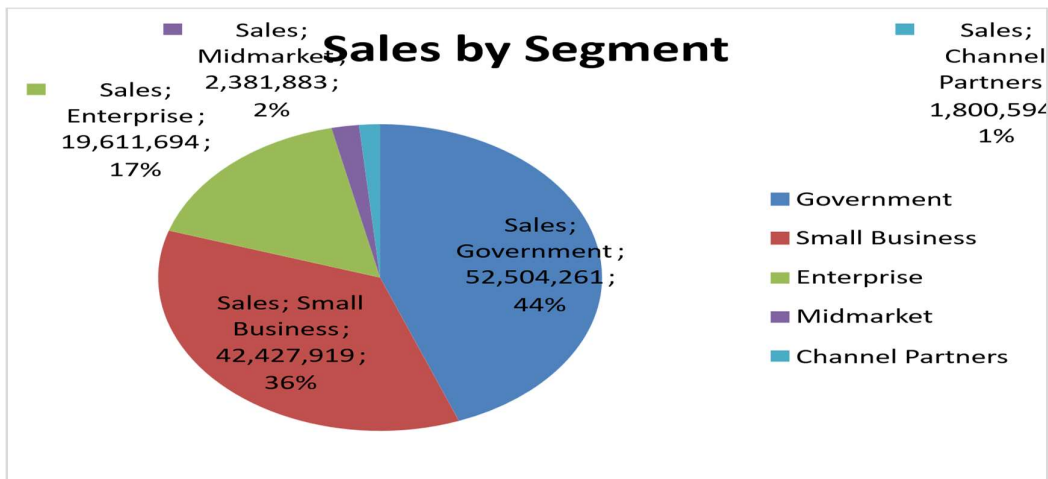


Figure 4

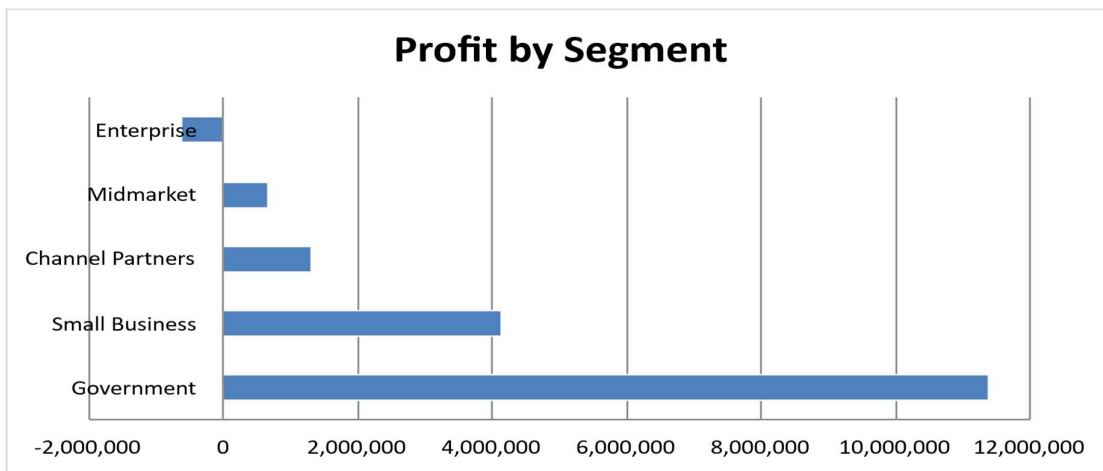


Figure 5



Figure 6

6. Detailed Findings

- **Overall performance:** The business generated Sales of 118.73M and Profit of 16.89M in 2013-2014, a healthy overall Profit Margin of 14.2%.
- **Strong year-over-year growth:** Sales nearly quadrupled from 26.4M in 2013 to 92.3M in 2014, indicating strong business momentum.
- **Country performance is balanced:** United States (25.0M), Canada (24.9M) and France (24.4M) are close leaders in sales; Mexico trails at 20.9M.
- **Paseo is the star product:** It contributes the highest profit (4.80M), roughly 58% more than the next best product, VTT (3.03M).
- **Government segment is the profit engine:** It alone contributes 11.39M in profit - over 67% of total company profit.
- **Enterprise segment is loss-making:** Despite generating sales, the Enterprise segment posted a net loss of about -0.61M, dragging down overall profitability.
- **Discounting has a clear tipping point:** Low (6.19M) and Medium (5.58M) discount bands generate the most profit, High discounts (3.39M) and No Discount (1.74M) generate less, suggesting moderate discounting is the sweet spot.
- **Product concentration risk:** The top two products (Paseo, VTT) account for roughly 46% of total profit, showing reliance on a small number of products.

7. Recommendations

- **Investigate the Enterprise segment loss:** Review pricing, discount levels and cost-to-serve for Enterprise accounts to identify why this segment is unprofitable, and consider renegotiating terms or reducing high-cost discounts there.
- **Double down on Paseo and VTT:** Increase marketing spend, inventory priority and sales focus on the top two profit-driving products.
- **Protect the Government relationship:** Since Government drives the majority of profit, prioritize account management and contract renewals for this segment.
- **Cap discount levels strategically:** Favor Low/Medium discount bands over High discounts, which erode margin without a proportional profit gain.
- **Grow the Mexico market:** Sales in Mexico lag other countries; targeted promotions or local partnerships could close this gap.
- **Diversify the product mix:** Reduce reliance on just two products by developing growth plans for Amarilla, Montana and Carretera.
- **Replicate 2014's growth drivers:** Analyze what changed between 2013 and 2014 (new customers, campaigns, pricing) and repeat those strategies going forward.

8. Deliverables & Conclusion

This case study demonstrates a complete, end-to-end data analytics workflow: cleaning raw data in Python, analyzing it with SQL/Pandas, visualizing it through Excel and Power BI dashboards, and compiling findings into this final report. The analysis shows a fundamentally healthy and growing business, with clear, actionable opportunities to improve profitability by addressing the Enterprise segment loss and optimizing the discount strategy.

Final Deliverables Submitted

- Clean_Explore_data.ipynb - Python data cleaning notebook
- Data_analysis_sql_pandas.ipynb - SQL/Pandas analysis notebook
- Sample_data.csv
- cleaned_data.csv Cleaned dataset
- End_to_End_Excel_Dashboards.xlsx - Excel workbook with dashboard and charts
- End_to_End_PowerBI_Dashboard.pbix - Power BI interactive dashboard
- This Final Report (PDF)